

Model-Free Deadbeat Predictive Current Control for SPMSM Based on Adaptive Gain Extended State Observer

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Abstract—The conventional model-free deadbeat predictive current control method with extended state observer (MFDPPC-ESO) has been a promising method because of its great dynamic performance and significant parameter robustness. However, the conventional extended state observer (ESO) still depends on inductance for designing controller gain, which inevitably reduces the parameter robustness of the method. In order to address this issue, a model-free deadbeat predictive current control (MFDPPC) method with adaptive gain ESO (AGESO) is proposed in this article. First, the impact of the conventional ESO input gain on control performance was analyzed, and it points out that controller gain mismatch will increase current ripple and even cause system instability. Then, the proposed method is based on a modified ultralocal model, in which the disturbance only includes the inductance term. Using disturbance as the objective function and combining finite-time gradient method (FGD) to estimate inductance parameters adaptively, the effect of controller gain mismatch is suppressed. Finally, experimental research was conducted on a 1-kW surface-mounted permanent magnet synchronous motor (SPMSM) test bench to verify the steady-state performance and parameter robustness of the method.

Index Terms—Deadbeat predictive current control (DPCC), parameter mismatch, robustness, surface-mounted permanent magnet synchronous motor (SPMSM).

I. INTRODUCTION

SURFACE-MOUNTED permanent magnet synchronous motors (SPMSMs) have high power density and a wide speed range. It can meet the requirements for motor performance in different fields [1], [2]. As an essential medium for developing industry, technology, and other industries today, the study of high-performance control methods for drive motors has become an important subject [3], [4], [5].

Deadbeat current predictive control (DPCC), as it achieves fast dynamic performance and excellent steady-state tracking performance with its low computational complexity, has found

application in various industrial fields [6], [7], [8]. However, the disadvantage of the DPCC algorithm is that it requires high accuracy of the motor dynamic model in practical operation [9]. In practical application, as the motor's operating temperature increases, the motor's resistance, inductance, and other parameters will change, leading to parameter mismatch, resulting in increased torque and current ripple [10], [11], [12]. Therefore, to enhance control performance and stability and improve the system's robustness to parameter variations [13], some researchers have developed different approaches. These approaches are mainly divided into three groups, including parameter identification [14], [15], [16], [17], [18], the disturbance observer method [19], [20], [21], [22], [23], [24], and model-free control (MFC) [25], [26], [27], [28].

Parameter identification is an important means to solve the problem of motor model parameters deviating from the original design values. At present, online identification methods include gradient descent (GD) [14], extended Kalman filter (EKF) [15], [16], and model reference-based adaptive systems (MRASs) [17], [18]. Lai et al. [14] proposed a method based on GD and q -axis harmonic voltage injection to identify the maximum torque angle. However, the introduction of harmonic signals into the q -axis inevitably deteriorates control performance and output torque. In [15], an inductance parameter identification method based on EKF is proposed. In [17], an enhanced DPCC method was proposed, which identifies the permanent magnet flux and stator inductance based on MRAS; nevertheless, it does not address the identification of the stator resistance. Compared with other similar algorithms, GD algorithms have a simpler structure and smaller computational complexity, which is beneficial for improving the dynamic response of the system. While effectively eliminating parameter mismatch and model uncertainty, the above methods overlook the nonlinear factors of the motor system, resulting in a particular impact on identification accuracy.

Another aspect is to introduce disturbance observers, which can compensate for disturbances caused by parameter mismatches and unmodeled disturbances [19]. In [20], a discrete Luenberger observer (LO) was proposed to estimate parameter disturbances and compensate for them, thereby enhancing the robustness of current control. LO is a linear observer with an intuitive structure, widely used in disturbance observer-based control. However, LO is sensitive to parameter changes, which may lead to inaccurate state estimation [21]. Thus, to improve the robustness of parameters, sliding mode observer (SMO) [22] and extended state observer (ESO) [23],

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[24] have become research hot spots. In [22], a high-order SMO is proposed, which effectively reduces chattering. Although SMO can achieve strong robustness, its design process is excessively repetitive, making it relatively difficult to determine its control coefficients. In contrast, ESO demonstrates strong robustness and easy implementation, making it more suitable for observing unknown disturbances compared to both LO and SMO.

Meanwhile, MFC has been widely studied as a strong robustness method. This method uses the input and output of the system to establish a data-driven model [25], replacing the physical model to describe the system state of the motor, decreasing the sensitivity to motor parameters. Tinazzi et al. [26] proposed a recursive least-squares (RLS) algorithm in the current prediction model, and this method recursively predicts the current value at the next moment based on the current samples from the previous periods. Lin et al. [27] established a lookup table (LUT) that stores the relationship between current changes corresponding to eight voltage vectors and combined this information with the predefined cost function to predict the optimal voltage vector the next time. Zhang et al. [28] approximated the controlled object as a first-order ultralocal model, representing the known and unknown parts as a single variable, and estimated and predicted this part and the state variable online through ESO. This method only requires a small amount of computation and can achieve strong robustness and good dynamic performance. However, the controller gain in the control algorithm is set to the reciprocal of the inductance, which means that the mismatch of inductance parameters will inevitably affect control performance. This is contrary to MFC's original intention.

In this article, a model-free deadbeat predictive current control (MFDPC) method with adaptive gain ESO (AGESO) method is proposed to solve the control performance degradation of the conventional ESO caused by controller gain mismatch. First, the transfer function between the reference current and the sampled current is obtained through z-transform, and the impact of the ESO input gain on control performance is analyzed. It points out that controller gain mismatch will increase the current ripple and even lead to system instability. Then, the conventional ultralocal model is modified to consider the rotor speed, resulting in a disturbance that includes only the inductance term. Leveraging the inductance information contained in the disturbance, a finite-time gradient method (FGD) is presented to update the controller gain adaptively. Finally, the experimental results show that compared to conventional ESO, the proposed method has better steady-state performance in the case of parameter mismatch.

II. CONVENTIONAL MFDPC WITH ESO

A. Conventional DPCC

In the dq -axis synchronous reference frame, neglecting the cross-saturation effect, eddy current losses, and iron losses, the stator voltage equations of SPMSM are as follows:

$$\begin{cases} \frac{di_d}{dt} = \frac{1}{L_s}u_d - \frac{R}{L_s}i_d + \omega_e i_q \\ \frac{di_q}{dt} = \frac{1}{L_s}u_q - \frac{R}{L_s}i_q - \omega_e i_d - \frac{\omega_e \psi_f}{L_s} \end{cases} \quad (1)$$

where i_d and i_q are the d -axis current and q -axis current, respectively, u_d and u_q are the d -axis and q -axis voltages, respectively, L_s is the SPMSM stator inductance, ψ_f is the permanent magnetic flux, and ω_e is the rotor electrical speed.

When the sampling period of the system is sufficiently short, the following relationship exists:

$$\begin{aligned} \frac{di_d}{dt} &= \frac{i_d(k+1) - i_d(k)}{T_s} \\ \frac{di_q}{dt} &= \frac{i_q(k+1) - i_q(k)}{T_s}. \end{aligned} \quad (2)$$

According to (2), by using the first-order forward Euler to discretize equation (1), the discrete state space function of SPMSM can be described as

$$i(k+1) = E(k)i(k) + Fu(k) + P(k) \quad (3)$$

where

$$\begin{aligned} i(k) &= \begin{bmatrix} i_d(k) \\ i_q(k) \end{bmatrix}, u(k) = \begin{bmatrix} u_d(k) \\ u_q(k) \end{bmatrix}, P(k) = \begin{bmatrix} 0 \\ -\frac{\psi_f}{L_s} T_s \omega_e(k) \end{bmatrix} \\ E(k) &= \begin{bmatrix} 1 - \frac{R}{L_s} T_s & T_s \omega_e(k) \\ -T_s \omega_e(k) & 1 - \frac{R}{L_s} T_s \end{bmatrix}, F = \begin{bmatrix} \frac{T_s}{L_s} & 0 \\ 0 & \frac{T_s}{L_s} \end{bmatrix}. \end{aligned}$$

Based on the DPCC principle, the reference voltage can be presented as

$$\begin{bmatrix} u_d^{\text{ref}}(k) \\ u_q^{\text{ref}}(k) \end{bmatrix} = F^{-1} \left\{ \begin{bmatrix} i_d^{\text{ref}}(k+1) \\ i_q^{\text{ref}}(k+1) \end{bmatrix} - E(k)i(k) - P(k) \right\} \quad (4)$$

where $i_d^{\text{ref}}(k+1)$ and $i_q^{\text{ref}}(k+1)$ are the d -axis and q -axis reference currents, respectively, representing the current that needs to be tracked next, and $u_d^{\text{ref}}(k)$ and $u_q^{\text{ref}}(k)$ are the voltages required to make the actual current of SPMSM accurately follow the reference value.

B. MFDPC With ESO

According to the mathematical model (3), the control performance of conventional DPCC is susceptible to the accuracy of the parameters used for the controller. Parameter mismatch will cause current ripple and steady-state current error [29].

Thus, to improve the parameter robustness of the DPCC, the ultralocal model is proposed to reconstruct the SPMSM mathematical model, and its expression is as follows:

$$\frac{di_{dq}}{dt} = \alpha_0 u_{dq} + F_{dq}. \quad (5)$$

According to the ultralocal model (5), the discrete-time form of the predictive SPMSM model can be described as

$$i_{dq}(k+1) = i_{dq}(k) + \alpha_0 T_s u_{dq}(k) - \alpha_0 T_s F_{dq}(k) \quad (6)$$

where α_0 is the inverse of initial inductance L_s , and $F_{dq}(k)$ is assumed to be the unknown part of the plant. According to (5), i_{dq} and F_{dq} are selected as state variables to establish ESO. The ESO can be constructed as follows:

$$\begin{cases} e_{rr} = \hat{i}_{dq} - i_{dq} \\ \hat{i}_{dq} = \hat{F}_{dq} + \alpha_c u_{dq} + \beta_1 e_{rr} \\ \hat{F}_{dq} = \beta_2 e_{rr} \end{cases} \quad (7)$$

where $\hat{i}_{dq}$ and $\hat{F}_{dq}$ are the estimation of i_{dq} and F_{dq} in (5), α_c is the gain of controller input, and β_1 and β_2 are the error feedback coefficients of ESO. Combining (5) and (7), the transfer function between the estimated disturbance $\hat{F}_{dq}$ and the actual disturbance F_{dq} is expressed as follows:

$$\frac{\hat{F}_{dq}(s)}{F_{dq}(s)} = \frac{-\beta_2}{s^2 - \beta_1 s - \beta_2}. \quad (8)$$

According to (8), the characteristic equation of ESO can be expressed as

$$s^2 - \beta_1 s - \beta_2 = 0. \quad (9)$$

To simplify the design process, the coefficient ω_b is used to design the two roots of the characteristic equation. The coefficient of ESO can be expressed as

$$\begin{cases} \beta_1 = -2\omega_b \\ \beta_2 = -\omega_b^2. \end{cases} \quad (10)$$

The design of ω_b will determine the dynamic performance and stability of the current loop. By designing the coefficients of the observer, the conventional ESO can achieve good control performance.

Due to the high sampling frequency in practical digital control systems, the forward Euler discretization method can be used to discretize (7). The discrete state space function can be expressed as follows:

$$\begin{cases} e_{rr} = \hat{i}_{dq}(k) - i_{dq}(k) \\ \hat{i}_{dq}(k+1) = \hat{i}_{dq}(k) + T_s(\hat{F}_{dq}(k) + \alpha_c u_{dq}(k) + \beta_1 e_{rr}(k)) \\ \hat{F}_{dq}(k+1) = \hat{F}_{dq}(k) + T_s \beta_2 e_{rr}(k). \end{cases} \quad (11)$$

According to the DPCC theory, the reference voltage value can be presented as

$$U_{dq}^{\text{ref}}(k) = \frac{1}{\alpha_c} \left[\frac{i_{dq}^{\text{ref}}(k+1) - \hat{i}_{dq}(k)}{T_s} - \hat{F}_{dq}(k) \right]. \quad (12)$$

The dq -axis voltage reference value can be calculated when $\hat{i}_{dq}$, $\hat{F}_{dq}$, and α_c values are accurately obtained, as described in (12). Fig. 1 illustrates the whole control diagram. In the implementation process of this algorithm, the stator resistance and flux parameters of the motor are no longer needed. However, the controller gain depends on inductance parameters for design, which means that controller gain mismatch will affect the performance of conventional ESO.

III. INFLUENCE OF CONTROLLER GAIN ON CONVENTIONAL ESO

By applying (12) to model (6), one can obtain the relationship between the sampled current and the reference currents in the presence of inaccurate controller gain as in (13), shown at

$$\begin{cases} i_d(k+1) = \frac{\alpha_o}{\alpha_c} i_d^{\text{ref}}(k+1) + \frac{-\Delta\alpha}{\alpha_c} i_d(k) + T_s \left(F_d - \frac{\alpha_o}{\alpha_c} \hat{F}_d(k) \right) \\ i_q(k+1) = \frac{\alpha_o}{\alpha_c} i_q^{\text{ref}}(k+1) + \frac{-\Delta\alpha}{\alpha_c} i_q(k) + T_s \left(F_d - \frac{\alpha_o}{\alpha_c} \hat{F}_d(k) \right) \end{cases} \quad (13)$$

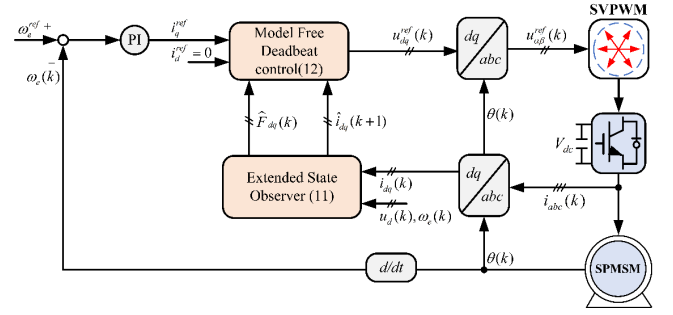


Fig. 1. Control block diagram of the conventional ESO.

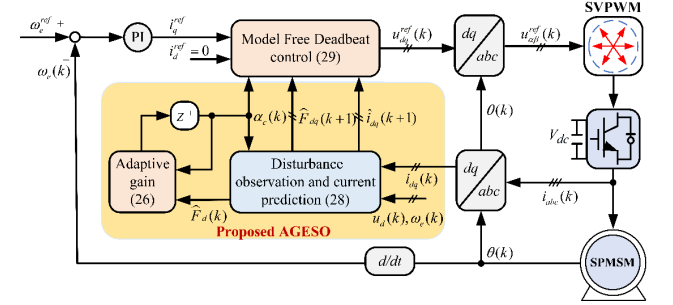


Fig. 2. Control block diagram of the proposed AGESO.

the bottom of the page, where $\Delta\alpha = \alpha_c - \alpha_o$ is the difference between the actual controller gain and the preset controller gain. It can be inferred from (13) that the conventional ESO has the following drawbacks.

1) The controller gain α_c mismatch will cause system instability. Since the mechanical time constant of the motor is much greater than the electrical time constant in the motor control system, the change in speed relative to current is very slow. Therefore, the term containing velocity ω_e in the disturbance F_{dq} changes very slowly in the current loop and can be regarded as an external disturbance. Taking i_{dq}^{ref} and i_{dq} as an output and input variable of the plant in (13) and considering $F_{dq}T_s - \frac{\alpha_o}{\alpha_c}T_s\hat{F}_{dq}(k)$ as external disturbance. Equation (13) in the z -domain can be expressed as

$$\frac{i_{dq}(z)}{i_{dq}^{\text{ref}}(z)} = \frac{(\alpha_o/\alpha_c)z}{z - (1 - \alpha_o/\alpha_c)}. \quad (14)$$

According to (14), the characteristic root of the system is $z = (1 - \alpha_o/\alpha_c)$. The necessary and sufficient condition for the stability of a discrete control system is that the poles of the system are within the unit circle in the z plane, and the stability conditions of the system can be obtained as follows:

$$-1 < 1 - \alpha_o/\alpha_c < 1. \quad (15)$$

From (15), it can be seen that the current loop will be unstable with the increment of α_o/α_c .

2) The mismatch of controller gain α_c will amplify the disturbance's high-frequency components and increase the current's ripple. The control performance of ESO depends on the accuracy of disturbance observations. When $\alpha_o/\alpha_c = 1$, in the steady state, (13) can be represented as

$$\begin{cases} i_d(k+1) = i_d^{\text{ref}}(k+1) + T_S(F_d(k) - \hat{F}_d(k)) \\ i_q(k+1) = i_q^{\text{ref}}(k+1) + T_S(F_q(k) - \hat{F}_q(k)). \end{cases} \quad (16)$$

It can be concluded that the DPCC is realized when ESO disturbance observations $\hat{F}_{dq}$ can accurately track actual disturbances F_{dq} . Compared to (5) and (1), the theoretical disturbance value is represented as follows:

$$\begin{cases} F_d = (\alpha_c - \alpha_0)u_d - \frac{R}{L_S}i_d + \omega_e i_q \\ F_q = (\alpha_c - \alpha_0)u_q - \frac{R}{L_S}i_q - \omega_e i_d - \frac{\psi_f}{L_S}\omega_e. \end{cases} \quad (17)$$

The first item in (17) is caused by a mismatched controller gain. Due to the nonlinearity of inverters, SVPWM modulation algorithms and overmodulation algorithms may generate higher order harmonic voltages. The mismatch of controller gain can amplify higher order voltage harmonics, leading to high-frequency disturbances. However, from (8), it can be seen that the conventional ESO is a second-order low-pass system. This indicates that under a fixed bandwidth value, the low-frequency signal in the disturbance F_{dq} will be accurately estimated, while the high-frequency signal will decay and cannot be accurately estimated, resulting in $\hat{F}_{dq} \neq F_{dq}$. This will inevitably deteriorate control performance, resulting in a larger current ripple.

Therefore, if the actual controller gain is inconsistent with the preset controller gain, it will directly deteriorate the control performance of the conventional ESO and even lead to system instability. In Section IV, an AGESO method is introduced to eliminate the negative impact caused by controller gain mismatch. The control block diagram of the proposed AGESO is shown in Fig. 2.

IV. PROPOSED MFDPPC WITH AGESO

A. Design of the AGESO

To facilitate controller gain adaptation, the improved ultralocal model of the SPMSM is presented as

$$\begin{cases} \frac{di_d}{dt} = \alpha_0 u_d + \omega_e i_q + F_d \\ \frac{di_q}{dt} = \alpha_0 u_q - \omega_e i_d + F_q \\ \frac{dF_{dq}}{dt} = G \end{cases} \quad (18)$$

Comparing (1) and (18), the theoretical disturbance value is

$$\begin{cases} F_d = (\alpha_c - \alpha_0)u_d - \frac{R}{L_S}i_d \\ F_q = (\alpha_c - \alpha_0)u_q - \frac{R}{L_S}i_q - \frac{\psi_f}{L_S}\omega_e. \end{cases} \quad (19)$$

The AGESO can be designed as follows:

$$\begin{cases} e_{rr} = \hat{i}_{dq} - i_{dq} \\ \frac{d\hat{i}_{dq}}{dt} = \mathbf{H}i_{dq} + \hat{F}_{dq} + \alpha_c u_{dq} + \beta_1 I e_{rr} \\ \frac{d\hat{F}_{dq}}{dt} = \beta_2 I e_{rr} \end{cases} \quad (20)$$

$$\text{where } \mathbf{H} = \begin{bmatrix} 0 & \omega_e \\ -\omega_e & 0 \end{bmatrix}.$$

From (19), it can be seen that the disturbance includes controller gain information, which means that controller gain parameter can be extracted from the disturbance through algorithms. In the proposed method, the control method with $i_d = 0$ is adopted. In the steady-state conditions, the disturbance observations can be expressed as

$$\hat{F}_d = F_d \approx (\alpha_c - \alpha_0)u_d. \quad (21)$$

In the presence of a controller gain deviation, (21) demonstrates that the d -axis disturbance deviates from zero. If the preset controller gain α_c is greater than the actual controller gain α_0 , the disturbance $\hat{F}_d$ will increase positively. Otherwise, the disturbance $\hat{F}_d$ will increase negatively.

B. Adaptation of α_c Based on FGD

As previously mentioned, the disturbance is 0 without controller gain mismatch. When the controller gain is mismatched, it will cause a significant change in the value of disturbance $\hat{F}_d$. Following this principle, a practicable α_c adaptation method combining ESO disturbance observations and FGD is proposed.

1) *Principle of FGD*: The implementation process of FGD is simple, as its calculation process follows the fast direction of GD, allowing the algorithm to achieve control requirements in a relatively short time. The FGD adopted in this article can be presented as [30]

$$\begin{cases} \dot{x} = -\theta \frac{\nabla f(x)}{\|\nabla f(x)\|} \\ \dot{\theta} = -\lambda\theta + \rho\|\nabla f(x)\| \end{cases} \quad (22)$$

where $f(x)$ is the objective function with an extreme point, θ is the search step size, and λ and ρ are the two coefficients used to adjust the adaptive iteration step size. Furthermore, it has been proven in [30] that by selecting appropriate coefficients, (22) can converge to the extreme point within a finite time. The convergence conditions and bounded time are expressed as follows:

$$\lambda^2 < 4\rho, \quad t \leq \frac{2\pi}{\sqrt{4\rho - \lambda^2}}. \quad (23)$$

2) *Adaptation of α_c* : Based on the principle of the FGD, if an objective function can be constructed for α_c and α_0 , then α_c can converge to α_0 within a finite time through gradients. In this article, the ESO disturbance observation $\hat{F}_d$ is used to construct an objective function without any motor parameter.

As shown in (21), the disturbance value is directly proportional to $\alpha_c - \alpha_0$. When α_c converges to α_0 , the disturbances $\hat{F}_d = 0$. Therefore, the objective function can be designed as follows:

$$f(\alpha_c) = (\hat{F}_d)^2 = [(\alpha_c - \alpha_0)u_d]^2 \quad (24)$$

and its gradient is as follows:

$$\nabla f(\alpha_c) = 2\hat{F}_d \frac{d[(\alpha_c - \alpha_0)u_d]}{dt} = 2\hat{F}_d u_d. \quad (25)$$

Because u_d varies a lot, the direct utilization of u_d in (25) will lead to significant fluctuations in the gradient value and affect the convergence of the controller gain. In addition, as a dc, the u_d value can be equivalent to a proportional gain of (25)

in stable conditions. Based on (20), u_d can be designed as the reciprocal of $\hat{\alpha}_c(u_d \approx p/\hat{\alpha}_c)$, which not only eliminates the negative impact of using u_d directly but also serves as a proportional gain function for (25). Thus, (25) is rewritten as

$$\nabla f(\alpha_c) = \frac{2p\hat{F}_d[k]}{\alpha_c[k]} \quad (26)$$

where $p > 0$ is a ratio coefficient, and its value determines the convergence speed. However, an excessive value of p will cause jitter. Therefore, there is a tradeoff between convergence speed and jitter, and p is set to 0.1 in this article.

Then, to be applied in practical systems, the discrete structure of (22) is represented as follows:

$$\begin{cases} \alpha_c(k+1) = \alpha_c(k) - T_s \theta(k) \frac{\nabla f(\alpha_c(k))}{\|\nabla f(\alpha_c(k))\|} \\ \theta(k+1) = (1 - T_s \lambda) \theta(k) + T_s \rho \|\nabla f(\alpha_c(k))\|. \end{cases} \quad (27)$$

In this article, to ensure the convergence of the controller gain α_c , the parameters λ and ρ are set to $2e4$ and $3e8$. Then, the gradient (26) is calculated based on the disturbance to achieve adaptive adjustment of the controller gain. It is worth noting that filters need to be used to smooth the results due to the presence of high-frequency signals.

3) *Discrete Structure of the AGESO*: To facilitate implementation on discrete systems, the formula (20) after forward Euler discretization is utilized to represent the following:

$$\begin{cases} e_{rr} = \hat{i}_{dq}(k) - i_{dq}(k) \\ \hat{i}_{dq}(k+1) = \mathbf{H}i_{dq}(k) + T_s(\hat{F}_{dq}(k) + \alpha_c(k)u_{dq}(k) + \beta_1 e_{rr}(k)) \\ \hat{F}_{dq}(k+1) = \hat{F}_{dq}(k) + T_s \beta_2 e_{rr}(k). \end{cases} \quad (28)$$

Based on the above equation, an accurate ultralocal current prediction model can be obtained based on the designed controller gain adaptive algorithm. Consequently, the proposed method has better parameter robustness compared to conventional ESO and can achieve better control performance.

C. Reference Voltage Vector Calculation

According to (18), combining the system disturbance observed by the ESO, the voltage reference vector can be presented as

$$U_{dq}^{\text{ref}}(k) = \frac{i_{dq}^{\text{ref}}(k+1) - i_{dq}(k)}{\alpha_c(k)T_s} - \frac{\hat{F}_{dq}(k) - \omega_e(k)i_{dq}(k)}{\alpha_c(k)}. \quad (29)$$

Among them, α_c can be updated in real time based on disturbances. The flowchart of the proposed method is depicted in Fig. 3. First, the algorithm observes the total disturbance of the system by sampling the motor current and speed. Then, based on the disturbance observations and FGD, the controller gain obtains real-time adaptive updates and is applied to the current prediction equation. Ultimately, the reference voltage is calculated using (29) and applied to the motor employing the SVPWM method.

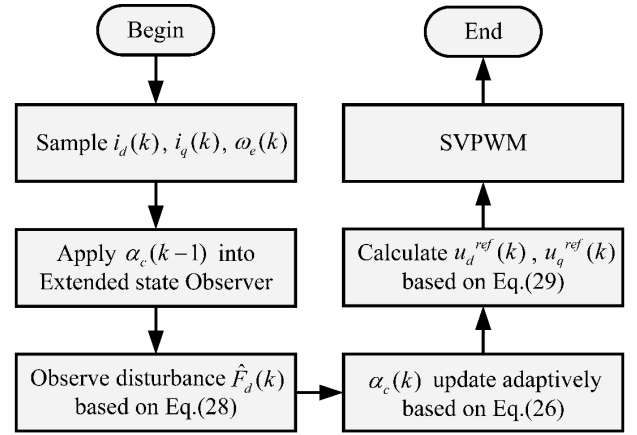


Fig. 3. Flowchart of the AGESO.

TABLE I
PARAMETERS OF THE SPMSM

Symbol	Parameters	Values
J	Rotational inertia	0.85 g·m ²
n	Rated speed	2500rpm
T _e	Rated torque	4 N·m
I _N	Rated current	4 A
n _p	Number of pole pairs	4
R _s	Stator resistance	1.7Ω
L _s	Stator inductance	3.21 mH
ψ _f	Rotor flux linkage	0.14Wb

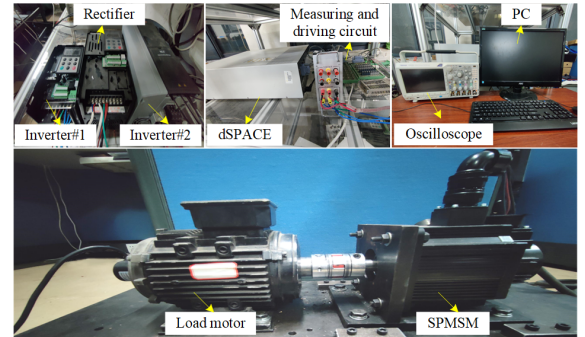


Fig. 4. Experimental platform of 1-kW SPMSM control system.

V. EXPERIMENTAL RESULTS

A. Test Bench Setup

In this section, the effectiveness of the proposed method is carried out on a 1-kW SPMSM system drive using a dSPACE controller. The experimental setup is shown in Fig. 4. In the speed control loop, the proportional and integral coefficients of the PI controller are set to 15 and 20, respectively. The main SPMSM parameters used in the experiment are listed in Table I. At the same time, the control frequency of the system is 10 kHz.

B. α_c Adaptive Performance Verification

In this section, the adaptive performance of the controller gain α_c is evaluated under various conditions. The experimental results are shown in Figs. 5–7. From the experimental results, it can be seen that regardless of the speed, load,

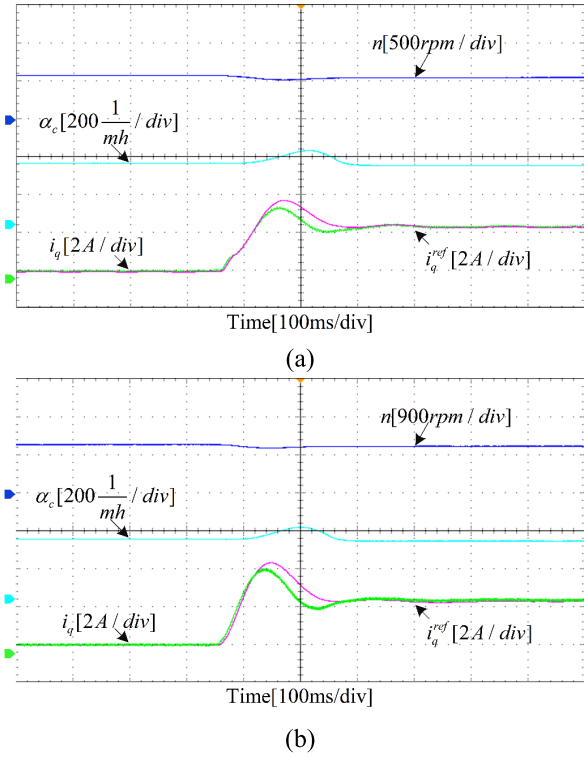


Fig. 5. α_c adaptive performance under different speed conditions. (a) 600 and (b) 1200 r/min.

or parameter mutation, the estimated value of the controller gain is approximately equal to the actual value of the reciprocal of the motor inductance ($\alpha_c \approx 312 \approx 1/L_0$), which proves the effectiveness of the control algorithm. The experiment waveforms include the α_c estimation value, speed, and the sampled and reference values of the dq -axis current.

1) *Adaption of α_c Against Load Changes*: Fig. 5 demonstrates that under different speed conditions, the controller gain can efficiently converge to the actual value in a relatively short time when the load is increased from no load to 40%. The proposed method shows outstanding anti-interference capability by sustaining its adaptability across different speed conditions, regardless of sudden changes in load torque.

2) *Adaption of α_c Against Various Speeds*: Fig. 6 shows the adaptive performance under the acceleration process of the motor. From the experimental results, although the α_c estimation value may fluctuate during acceleration, it is still accurately estimated when the speed stabilizes.

3) *Adaption of α_c Against Parameter Changes*: In Fig. 7, the adaptive performance is demonstrated under no-load conditions with sudden changes in the gain parameters during operation. The conducted experiments reveal that the controller gain value can rapidly converge adaptively to the actual value, enabling the sampled currents i_q to accurately track the references i_q^{ref} . Consequently, the current error caused by parameter mismatch can be swiftly eliminated.

C. Comparative Study

To validate the superior robustness of the proposed method, this section compares the proposed AGESO with

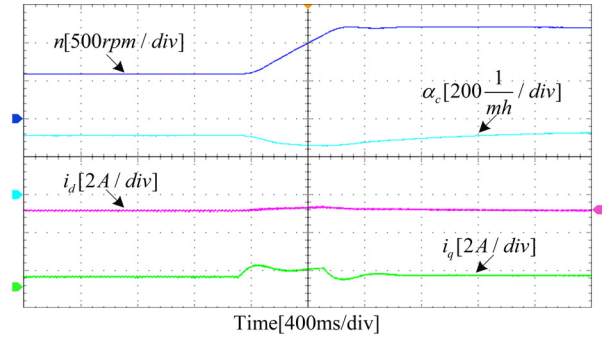


Fig. 6. α_c adaptive performance under speed sudden changes from 600 to 1200 r/min.

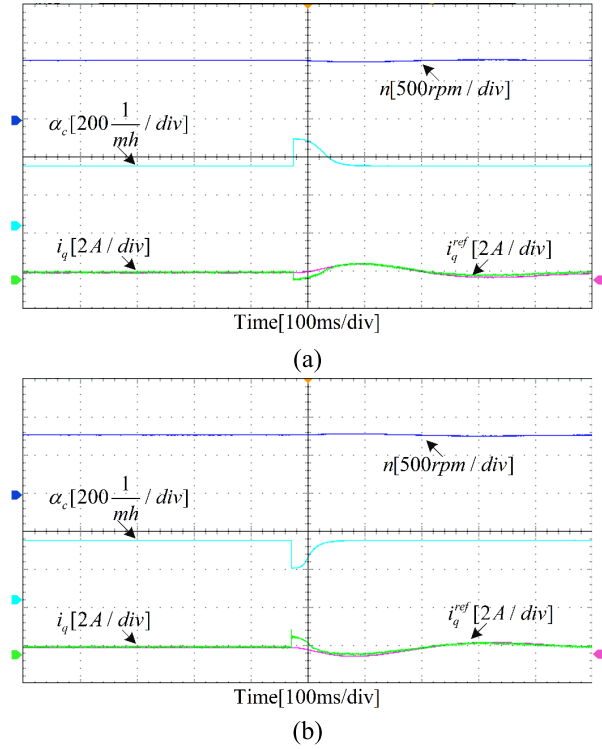


Fig. 7. α_c adaptive performance under gain parameter sudden change at 800 r/min. (a) Gain parameter suddenly increases 50%. (b) Gain parameter suddenly decreases 100%.

the conventional ESO. To ensure fairness in comparison, the bandwidth ω_b of both methods is set to 100 rad/s. The experimental results are presented in Figs. 8 and 9.

1) *Transient Performance*: The comparison in Fig. 8 illustrates the performance difference in current transients when a q -axis current step is applied at 800 r/min under a $0.5 L_S$ inductance mismatch condition. It is evident that the conventional ESO with mismatched controller parameters results in significant current ripples. Differently, the adoption of the proposed method can lead to a reduction in current ripples. Furthermore, the dynamic current tracking performance of AGESO surpasses that of conventional ESO.

2) *Steady-State Performance*: Fig. 9 depicts the steady-state experimental comparison results of the motor at 600-r/min speed under $0.3 L_S$ inductance mismatch condition. Compared to Fig. 9(a) and (b), the proposed method has a smaller

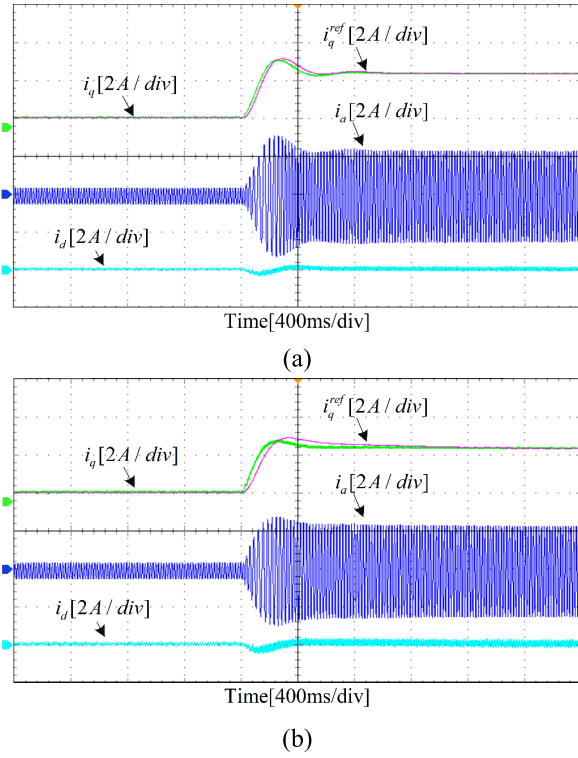


Fig. 8. Dynamic experimental results with load torque change (0%–40%) at the speed of 800 r/min under the 0.5 L_S inductance mismatch. (a) Proposed AGESO. (b) Conventional ESO.

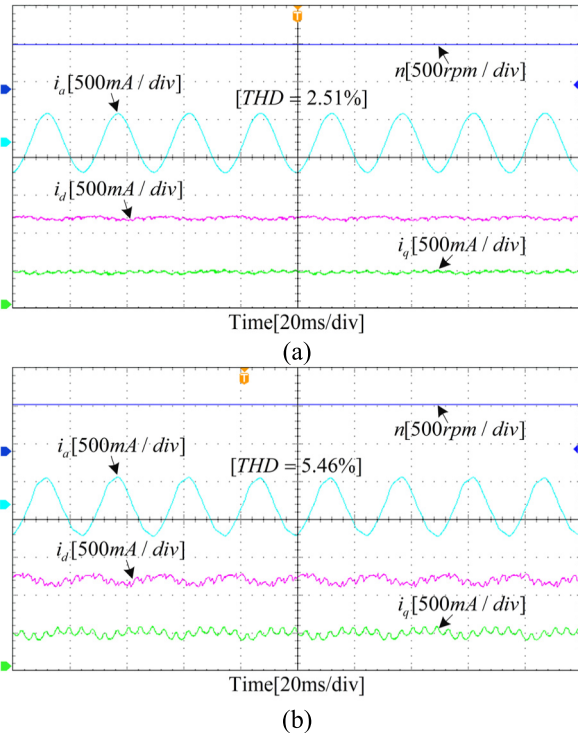


Fig. 9. Steady-state experimental results at the speed of 600 r/min under the 0.3 L_S inductance mismatch. (a) Proposed AGESO. (b) Conventional ESO.

dq -axis current ripple under the steady-state conditions. At the same time, the total harmonic distortion of the conventional ESO control method and the proposed method is 5.46% and 2.51%, respectively. This means that the proposed method

can adaptively update the controller gain, avoid parameter mismatch, and obtain a better steady-state current waveform.

VI. CONCLUSION

This article proposes an MFDPPC method with AGESO and performs experimental verification in the SPMSM system. The primary contributions of this article are as follows: 1) analyzing the impact of initial controller gain mismatch on ESO control performance and pointing out that controller gain mismatch will result in increased current ripple and even system instability and 2) based on a modified ultralocal model, an AGESO method is proposed, which utilizes the inductance information contained in disturbances and combines FGD to update the controller gain adaptively, thereby a strong robustness control based on MFDPPC has been achieved.

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